

1. WAFT Typst Infrastructure - Complete System

Production-Ready Document Generation System

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1.1. System Overview

The WAFT Typst infrastructure is a comprehensive, production-ready system for generating PDF documents using Typst. The system includes:

- **Secure Compilation Engine:** TypstCompiler with comprehensive security hardening
- **Auto-Discovery Registry:** TypstTemplateRegistry with metadata extraction
- **11 Templates:** Covering academic, business, and creative use cases
- **Unified API:** Consistent interface across all templates

1.2. Template Catalog

1.2.1. Academic Templates (3 templates)

Charged Ieee (paper)

- Description: Charged IEEE Typst Template Wrapper
===== Python ...
- Tags: typst, academic, ieee, conference, paper

Unequivocal Ams (paper)

- Description: Unequivocal AMS Typst Template Wrapper
===== P...
- Tags: typst, academic, ams, mathematics, paper

Arkheion (preprint)

- Description: Arkheion Typst Template Wrapper =====
Python wrapper ...
- Tags: typst, academic, arxiv, preprint, paper

1.2.2. Business Templates

Appreciated Letter (letter)

- Description: Appreciated Letter Typst Template Wrapper
=====...

Dashing Dept News (newsletter)

- Description: Dashing Dept News Typst Template Wrapper
=====...

Flow Way (report)

- Description: Flow-Way Typst Template Wrapper =====
Python wrapper ...

1.2.3. Creative Templates

Wonderous Book (book)

- Description: Wonderous Book Typst Template Wrapper
===== Pyt...

Badformer (game)

- Description: Badformer Typst Template Wrapper =====
Python wrapp...

Cereal Words (game)

- Description: Cereal Words Typst Template Wrapper
===== Python ...

Icicle (game)

- Description: Icicle Typst Template Wrapper ===== Python wrapper for ...

1.3. System Statistics

- Total Templates: 11
- Categories: 8
- Unique Tags: 30
- Security Features: 4 (path validation, size limits, timeout, subprocess security)
- Test Coverage: 43 tests (all passing)

1.4. Key Features

1.4.1. Security

- Path validation prevents directory traversal
- Content size limits (default 10MB, configurable)
- Compilation timeout (default 60s, configurable)
- Subprocess security (shell=False, list arguments)

1.4.2. Performance

- Single-pass compilation (faster than LaTeX)
- Typical compilation: < 1 second
- Efficient resource usage
- Automatic cleanup

1.4.3. Developer Experience

- Auto-discovery of templates
- Clear error messages
- Comprehensive documentation
- Unified API across templates

1.5. Usage

```
from src.waft.templates.typst import get_typst_registry

registry = get_typst_registry()
generate = registry.get_generate_function("Flow Way")

pdf = generate(
    title="My Document",
    content="# Content",
    output_path=Path("output.pdf")
)
```

1.6. Integration

The infrastructure integrates seamlessly with:

- WAFT DocumentBuilder system
- Existing LaTeX template infrastructure
- D&D 5e content generation (planned)

- Automated document pipelines

2. Conclusion

The Typst infrastructure is production-ready and provides a modern, secure, and efficient alternative to traditional document generation systems. With comprehensive template support, security hardening, and excellent performance, it's ready for use in production environments.

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Generated by WAFT Typst Infrastructure